

phas1_SLR

June 14, 2026

```
[2]: import importlib
import pandas as pd
import numpy as np
import re
from pathlib import Path

import functions
importlib.reload(functions)
from functions import *

df_1111 = pd.read_csv("datasets/AI_Lit_Que_1111.csv")
df_1204 = pd.read_csv("datasets/AI_Lit_Que_1204.csv")

# add source/course label
df_1111["course"] = "1111"
df_1204["course"] = "1204"

print(df_1111.shape, df_1204.shape)
```

(118, 37) (23, 37)

1 EDA

1.1 Descriptive statistics

1.1.1 Mean comparison

SES

- Class 1111 is very slightly below the pooled SES average
- Class 1204 is slightly above the pooled SES average

The difference is present but not large in raw standardized units.

- $Skew < 0$: SES is somewhat left-skewed, especially in 1204. More observations are clustered on the higher side, with a tail toward lower SES-index values.

AI Literacy

- Class 1204 appears to have higher AI literacy on average than class 1111. The gap here is much more visible than for SES.

```
[3]: df, meta = prepare_dataset("AI_Lit_Que_1111.csv", "AI_Lit_Que_1204.csv")
```

```
# prepare_dataset is a preprocessing function defined in functions.py
construct_summary_table(df)
```

```
-----
FileNotFoundError                                Traceback (most recent call last)
<ipython-input-3-fd06e069d22a> in <module>
----> 1 df, meta = prepare_dataset("AI_Lit_Que_1111.csv", "AI_Lit_Que_1204.csv")
      2
      3 # prepare_dataset is a preprocessing function defined in functions.py
      4 construct_summary_table(df)

~/Documents/HKU/RA/GRF26/1st_analysis/functions.py in prepare_dataset(path_1111,
↳ path_1204)
    216     meta : dict                - lists of column names for each construct
    217         """
--> 218     d1 = pd.read_csv(path_1111).assign(course="1111")
    219     d2 = pd.read_csv(path_1204).assign(course="1204")
    220

~/Documents/anaconda3/lib/python3.8/site-packages/pandas/util/_decorators.py in
↳ wrapper(*args, **kwargs)
    209         else:
    210             kwargs[new_arg_name] = new_arg_value
--> 211         return func(*args, **kwargs)
    212
    213         return cast(F, wrapper)

~/Documents/anaconda3/lib/python3.8/site-packages/pandas/util/_decorators.py in
↳ wrapper(*args, **kwargs)
    329         stacklevel=find_stack_level(),
    330     )
--> 331     return func(*args, **kwargs)
    332
    333     # error: "Callable[[VarArg(Any), KwArg(Any)], Any]" has no

~/Documents/anaconda3/lib/python3.8/site-packages/pandas/io/parsers/readers.py
↳ in read_csv(filepath_or_buffer, sep, delimiter, header, names, index_col,
↳ usecols, squeeze, prefix, mangle_dupe_cols, dtype, engine, converters,
↳ true_values, false_values, skipinitialspace, skiprows, skipfooter, nrows,
↳ na_values, keep_default_na, na_filter, verbose, skip_blank_lines, parse_dates,
↳ infer_datetime_format, keep_date_col, date_parser, dayfirst, cache_dates,
↳ iterator, chunksize, compression, thousands, decimal, lineterminator,
↳ quotechar, quoting, doublequote, escapechar, comment, encoding,
↳ encoding_errors, dialect, error_bad_lines, warn_bad_lines, on_bad_lines,
↳ delim_whitespace, low_memory, memory_map, float_precision, storage_options)
```

```

948     kwds.update(kwds_defaults)
949
--> 950     return _read(filepath_or_buffer, kwds)
951
952

~/Documents/anaconda3/lib/python3.8/site-packages/pandas/io/parsers/readers.py
↳ in _read(filepath_or_buffer, kwds)
603
604     # Create the parser.
--> 605     parser = TextFileReader(filepath_or_buffer, **kwds)
606
607     if chunksize or iterator:

~/Documents/anaconda3/lib/python3.8/site-packages/pandas/io/parsers/readers.py
↳ in __init__(self, f, engine, **kwds)
1440
1441         self.handles: IOHandles | None = None
-> 1442         self._engine = self._make_engine(f, self.engine)
1443
1444         def close(self) -> None:

~/Documents/anaconda3/lib/python3.8/site-packages/pandas/io/parsers/readers.py
↳ in _make_engine(self, f, engine)
1733             if "b" not in mode:
1734                 mode += "b"
-> 1735             self.handles = get_handle(

1736                 f,
1737                 mode,

~/Documents/anaconda3/lib/python3.8/site-packages/pandas/io/common.py in
↳ get_handle(path_or_buf, mode, encoding, compression, memory_map, is_text,
↳ errors, storage_options)
854         if ioargs.encoding and "b" not in ioargs.mode:
855             # Encoding
--> 856             handle = open(

857                 handle,
858                 ioargs.mode,

FileNotFoundError: [Errno 2] No such file or directory: 'AI_Lit_Queue_1111.csv'

```

1.1.2 Measurement consistency

SES

- SES shows **moderate** consistency across both classes (pooled Cronbach's $\alpha = 0.69$).

AI Literacy

- AI literacy shows **low** consistency across both classes (pooled Cronbach's $\alpha = 0.52$; even lower in 1111).

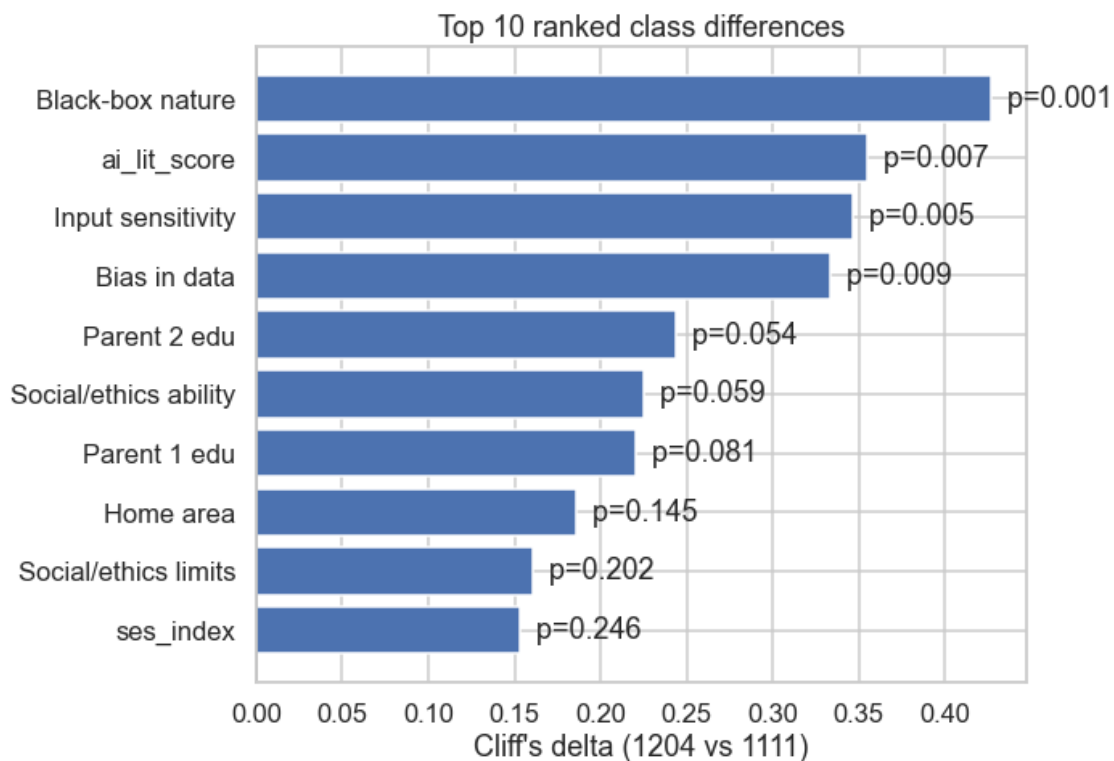
The AI literacy items do not appear to function as a highly homogeneous single scale in this sample.

```
[ ]: reliability_table(df, meta)
```

```
[ ]:      sample      construct      alpha  n_complete
0      1111  SES index ingredients  0.689986         118
1      1111    AI literacy score  0.466021         118
2      1204  SES index ingredients  0.739366          23
3      1204    AI literacy score  0.624770          23
4 Combined  SES index ingredients  0.693905         141
5 Combined    AI literacy score  0.523376         141
```

1.1.3 Effect sizes: Two-sided Mann–Whitney U test

```
[ ]: eff = effect_size_table(df, meta)
     plot_ranked_effect_sizes(eff, top_n=10)
```



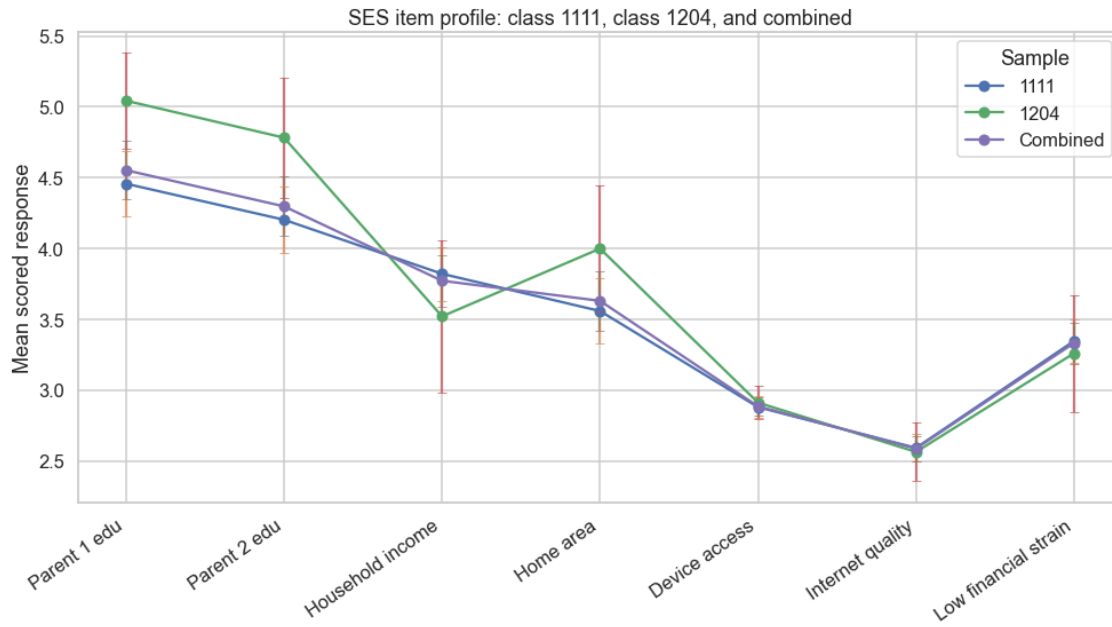
- Class 1204 tends to score higher than class 1111 on several AI literacy components, especially the more conceptual items.

- By contrast, the SES differences are weaker and mostly not statistically convincing.

1.2 Visualizations

1.2.1 Item profiles

```
[ ]: plot_item_profile(df, meta, construct="SES")
```



```
[ ]: plot_item_profile(df, meta, construct="AI")
```



1.2.2 Distribution of composite scores

```
[ ]: plot_ses_ai_correlation_heatmaps(df, meta)
```



- In the pooled sample, most correlations are weak to modest, with a tendency toward small positive associations rather than a single dominant pattern.
- The 1204 panel shows more extreme values, but given the much smaller subgroup size, those correlations are likely less stable.

2 SES Effect on AI Literacy (No mediation)

$$\text{AI literacy} = \beta_0 + \beta_1 \text{SES} + \varepsilon$$

```
[ ]: run_total_effect_models(df, ses_cols="ses_index", ai_cols="ai_lit_score")
```

```
[ ]:      sample ses_dimension  ai_outcome  beta_ses_std  p_hc3  ci_low_95  \
0  Combined      ses_index  ai_lit_score      0.195893   0.01   0.046838

      ci_high_95  r_squared  spearman_rho  spearman_p
0      0.344948   0.038374      0.201231   0.016723
```

- In the **combined sample**, SES shows a small but statistically supported positive association with AI literacy. Students with higher SES tend to have slightly higher AI literacy scores.
- In **class 1111**, the estimated SES effect is again positive and modest, and the result is statistically significant.
- In **class 1204**, the estimated effect is also positive, but highly uncertain and not statistically supported — most likely due to the much smaller sample size rather than evidence that no relationship exists.

The effect size is not large. In both 1111 and the combined sample, a one standard deviation increase in SES is associated with only about a 0.2 SD increase in AI literacy.

The explained variance is low ($R^2 = 0.04$), meaning SES accounts for only a small part of the variation in AI literacy. SES appears to matter, but is clearly not the only factor.

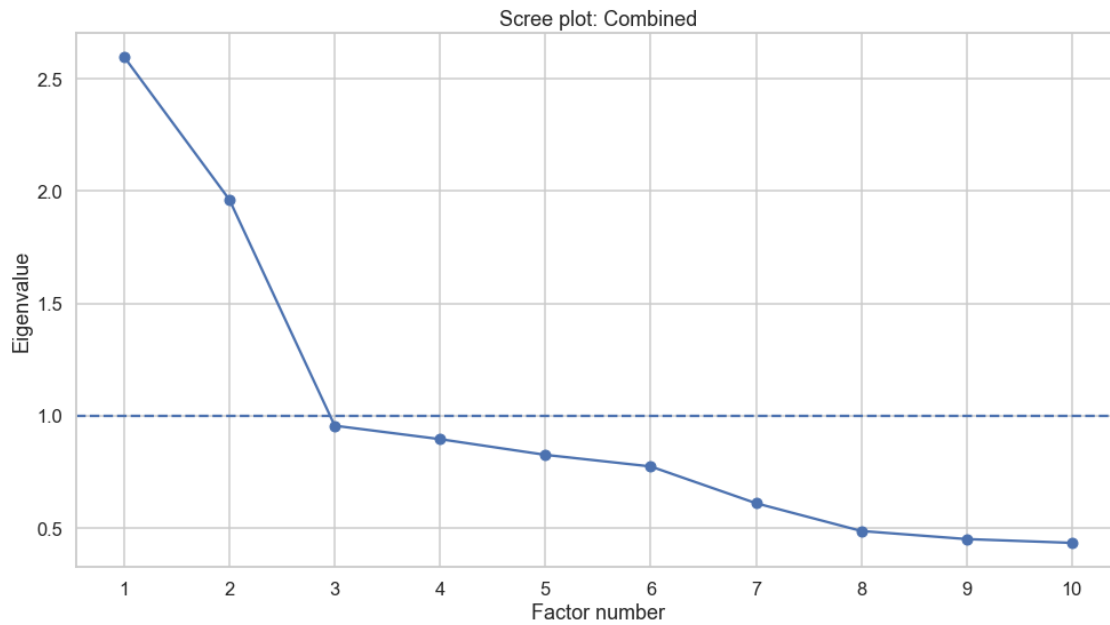
- The Spearman correlations are also positive in 1111 and in the combined sample, supporting the same conclusion from a rank-based perspective.

3 Exploratory Factor Analysis (EFA)

3.1 EFA for AI Literacy

```
[ ]: ai_efa_items = [
    "ai_concept_data_bias_scored_num",
    "ai_concept_blackbox_scored_num",
    "ai_concept_input_variation_scored_num",
    "ai_concept_prompt_wording_scored_num",
    "ai_concept_social_ethics_scored_num",
    "ai_ability_training_data_scored_num",
    "ai_ability_explainability_scored_num",
    "ai_ability_input_sensitivity_scored_num",
    "ai_ability_prompting_scored_num",
    "ai_ability_social_ethics_scored_num",
]

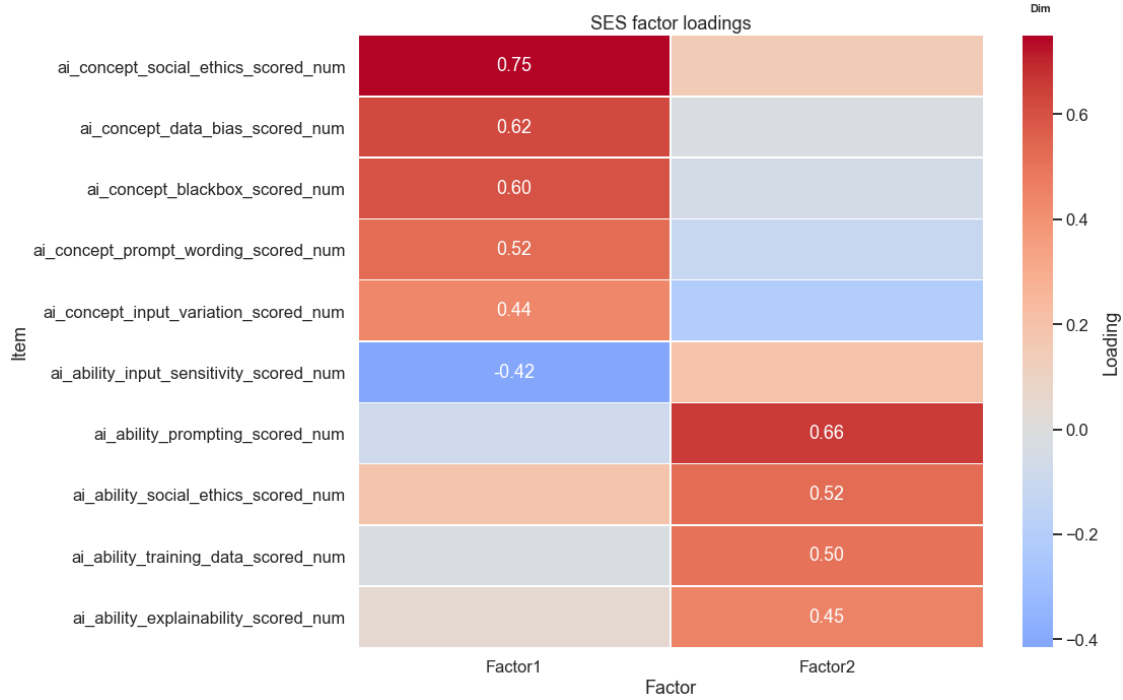
df_ai = df.copy()
diag_combined, kmo_items_combined = efa_diagnostics(df_ai, "Combined", ai_efa_items)
eig_combined = run_scree_analysis(df_ai, "Combined", ai_efa_items)
eig_combined
```



- The eigenvalues support a two-factor modeling. 2 principal components have eigenvalues > 1.

```
[ ]: fa_combined, d_combined, loadings_combined, variance_combined = fit_efa(
    df=df_ai,
    sample="Combined",
    items=ai_efa_items,
    n_factors=2,
    rotation="oblimin"
)

plot_loading_heatmap(loadings_combined, cutoff=0.30, title="SES factor_1
↳loadings")
```

- *Factor 1 is dominated by the conceptual items:* data bias, black-box nature, input variation, prompt wording, and especially social/ethics limits all load positively on this factor.

That means Factor 1 can be interpreted as something like conceptual AI understanding

- *Factor 2 is dominated by the self-reported competence items:* explaining training data, explainability, prompting ability, and social/ethics ability load on this factor.

Factor 2 looks like self-reported AI ability/confidence.

3.2 SES variables

3.2.1 Special test for ‘other’ school vs non-other school

- Two sided t-test:

$$H_a : \mu_{\text{other school}} \neq \mu_{\text{non-other school}}$$

```
[ ]: school_others_ses_ttest(df).round(3)
```

```
[ ]:
      metric  n_others  mean_diff_others_minus_nonothers  t_stat  \
0  Parent 2 education      17                1.066    4.082
1  Parent 1 education      17                0.909    3.422
2      Home area          17                0.486    1.769
3      SES index          17                0.198    1.425
4  Device access          17               -0.205   -1.405
```

5	Low financial strain	17	-0.045	-0.202
6	Household income	17	-0.009	-0.032
7	Internet quality	17	-0.000	-0.003

	p-value	cohens_d
0	0.000	0.866
1	0.002	0.753
2	0.090	0.387
3	0.169	0.348
4	0.177	-0.522
5	0.842	-0.050
6	0.975	-0.009
7	0.998	-0.001

- Conclusion:

There is no significant difference between ‘other school’ and ‘non-other school’ in terms of most SES metrics except Parent education.

- *Assigns the ‘Others school’ group to the known group with the smallest Euclidean distance*

Uses the students in the 3 known school groups as reference. Computes the centroid (mean SES profile) for each known group and the centroid for the ‘Others school’ group.

3.2.2 EFA for SES

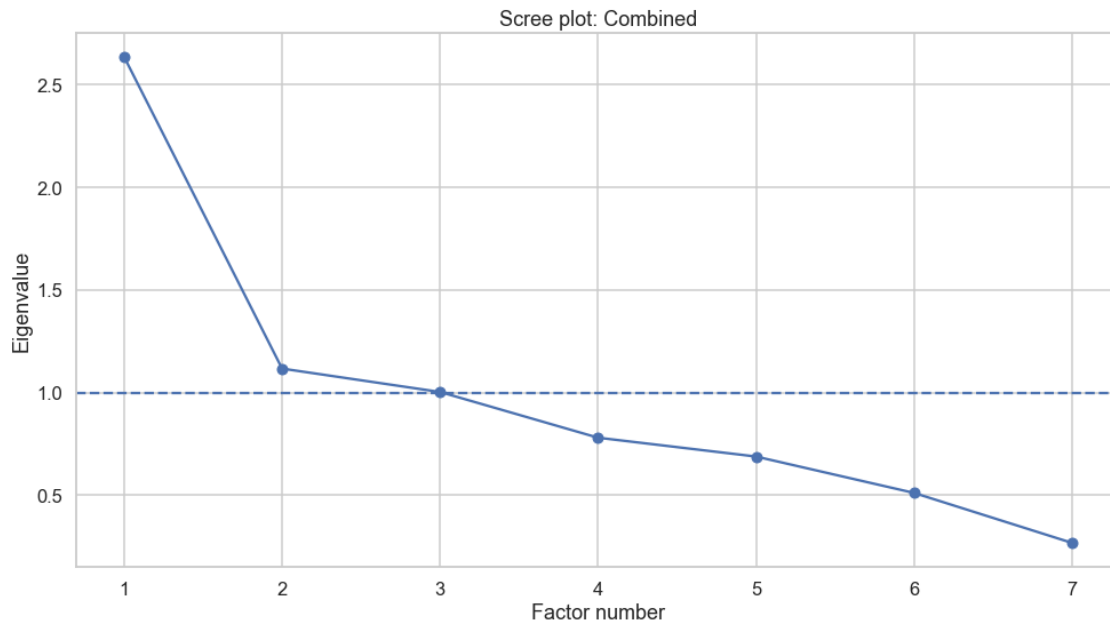
- new feature: Space_per_person
- quantified school type and housing type

```
[ ]: # Language SES is not included here

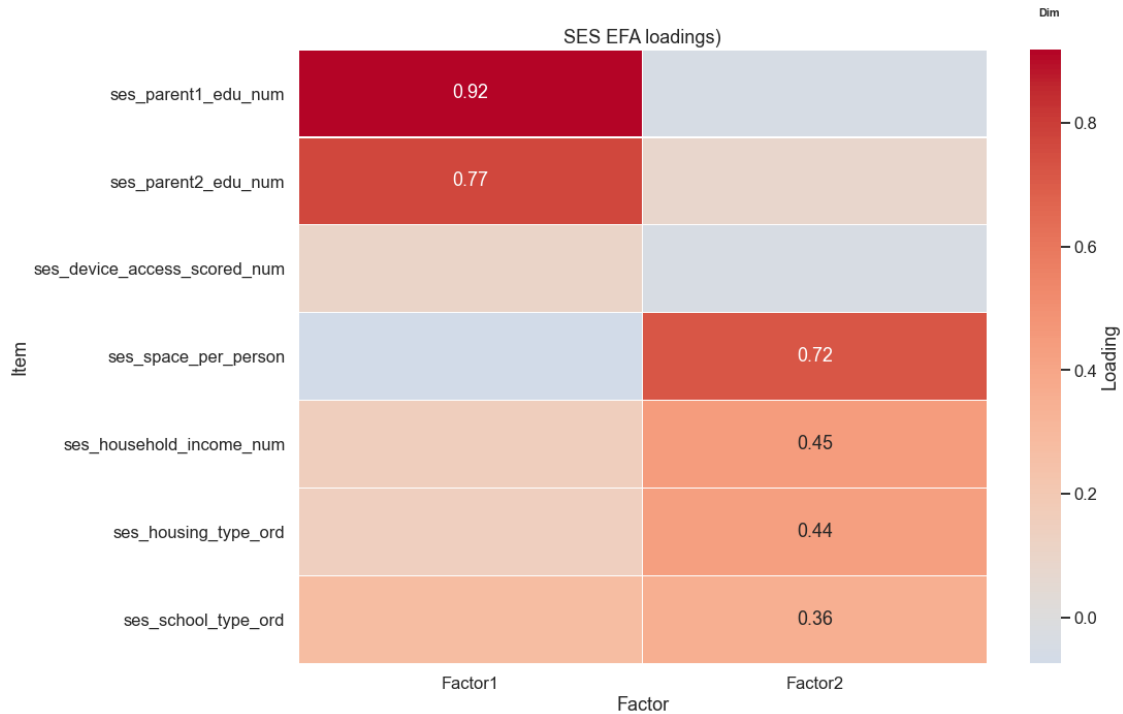
df_ses = add_ses_space_house_variables(df)

ses_efa_items_2dim = [
    "ses_parent1_edu_num",
    "ses_parent2_edu_num",
    "ses_device_access_scored_num",
    "ses_household_income_num",
    # "ses_financial_constraint_scored_num",
    "ses_space_per_person",
    # "ses_household_size_num",
    "ses_housing_type_ord",
    # "ses_internet_quality_scored_num",
    "ses_school_type_ord",
]

diag_ses_combined, kmo_ses_combined = efa_diagnostics(df_ses, "Combined",
↪ses_efa_items_2dim)
eig_ses_combined = run_scee_analysis(df_ses, "Combined", ses_efa_items_2dim)
```



```
[ ]: fa_ses2, d_ses2, loadings_ses2, variance_ses2 = fit_efa(  
    df=df_ses,  
    sample="Combined",  
    items=ses_efa_items_2dim,  
    n_factors=2,  
    rotation="oblimin"  
)  
  
plot_loading_heatmap(  
    loadings_ses2,  
    cutoff=0.3,  
    title="SES EFA loadings")  
)
```



- We may probably try two dimensions in SES

Conclusion

- Based on the EFA, SES can be treated as two dimensions:
 1. Family / institutional background ses_parent1_edu_num ses_parent2_edu_num ses_school_type_ord
 2. Material resources ses_household_income_num ses_financial_constraint_scored_num ses_space_per_person ses_housing_type_ord

AI literacy is better represented by two related but distinct components: 1. Conceptual understanding

2. Ability/confidence.

- Refitting the simple linear model using the factor scores: It returns a disappointing result: the SES factor is not significantly associated with either AI factor, and the effect sizes are very small.

4 Mediator analysis

4.1 Mediator pairs

Each mediator is measured by two different questions. We should check the consistency of these pairs before using them in mediation analyses.

```
[ ]: mediator_reliability = mediator_reliability_table(df, mediator_map)
mediator_reliability
```

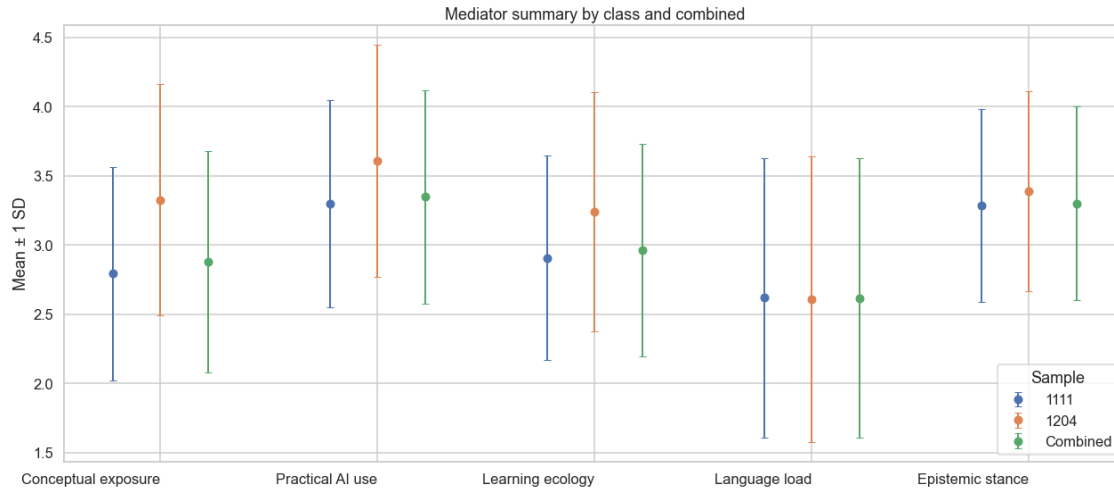
```
[ ]:
      sample      mediator  n_complete  spearman_rho  alpha_2item
0      1111  conceptual_exposure      118      0.292140      0.486054
1      1111    practical_ai_use      118      0.223236      0.379432
2      1111    learning_ecology      118      0.147209      0.274513
3      1111    language_load      118      0.601679      0.759845
4      1111    epistemic_stance      118      0.320206      0.564601
5      1204  conceptual_exposure       23      0.392293      0.567944
6      1204    practical_ai_use       23      0.367887      0.528373
7      1204    learning_ecology       23      0.399625      0.622269
8      1204    language_load       23      0.625154      0.779258
9      1204    epistemic_stance       23      0.360713      0.406300
10  Combined  conceptual_exposure      141      0.322244      0.522657
11  Combined    practical_ai_use      141      0.256532      0.419903
12  Combined    learning_ecology      141      0.184561      0.350871
13  Combined    language_load      141      0.603460      0.761378
14  Combined    epistemic_stance      141      0.334422      0.539159
```

- $\rho \geq 0.30$, $\alpha > 0.5$ means acceptable for a 2-item pair
- Most pairs show at least a modest positive association (by ρ and Alpha 2)
- The weakest pair is learning_ecology ($\rho = 0.18$, $p = 0.028$). This is still positive, but the association is fairly small. That suggests the two items may not be capturing exactly the same thing, even if they belong to the same broad domain.

Add composite mediator scores (averaging two items)

```
[ ]: df_med = add_mediator_composites(df, mediator_map)

# mediators table
med_summary = mediator_summary_table(df_med)
plot_mediator_summary(med_summary, errorbar="sd")
```



4.2 Indirect Effect Model specification

- Mediation models (a path)

$$\text{Mediator} = \alpha_0 + a \cdot \text{SES} + \varepsilon_M$$

- Outcome model (b path)

$$\text{AI literacy} = \beta_0 + c' \cdot \text{SES} + b \cdot \text{Mediator} + \varepsilon_Y$$

- Indirect effect ($a \cdot b$): greater than zero when a and b are both positive (or both negative)
- Total effect ($c = c' + a \cdot b$)
- Indirect effect coefficient and confidence interval are estimated using bootstrapping
- 2 * SES -> 5* mediators -> 2 * AI factors = 20 mediation models in total

4.3 Direct effect of selected split points of SES on AI literacy

```
[ ]: df_analysis = add_ses_ai_factor_scores(
    df_base=df_med,
    ai_df=df,
    ai_items=ai_efa_items,
    ai_fa=fa_combined,
    ses_df=df_ses,
    ses_items=ses_efa_items_2dim,
    ses_fa=fa_ses2
)

mediation_results_2ses = run_all_simple_mediations(
    df_analysis,
```

```

x=["ses_factor1_score", "ses_factor2_score"],
mediators=mediator_vars,
outcomes=["ai_factor1_score", "ai_factor2_score"],
sample_list=["Combined"],
n_boot=3000,
seed=2026
)

mediation_results_2ses["ses_dimension"] =_
↳mediation_results_2ses["ses_dimension"].map(ses_label_map).
↳fillna(mediation_results_2ses["ses_dimension"])
mediation_results_2ses["outcome"] = mediation_results_2ses["outcome"].
↳map(outcome_label_map).fillna(mediation_results_2ses["outcome"])
mediation_results_2ses["mediator"] = mediation_results_2ses["mediator"].
↳map(mediator_label_map).fillna(mediation_results_2ses["mediator"])
mediation_results_2ses.to_csv("mediation_results_2ses.csv", index=False)

```

4.3.1 Multiple testing correction

```

[ ]: mediation_results_2ses["p_fdr"] = multipletests(
    mediation_results_2ses["p_indirect_boot"], method="fdr_bh")

# mediation_results_2ses["p_fdr"]

```

```

[ ]:
    sample      ses_dimension      ai_outcome  beta_ses_std    p_hc3  \
0  Combined  ses_factor2_score  ai_factor1_score    0.060515  0.518543

    ci_low_95  ci_high_95  r_squared  spearman_rho  spearman_p
0  -0.123204   0.244234   0.003662    0.091949    0.278179

```

4.3.2 The indirect-effect forest plot

- The a path is SES -> Mediator_i
- The b path is the Mediator_i -> AI literacy while controlling for SES.

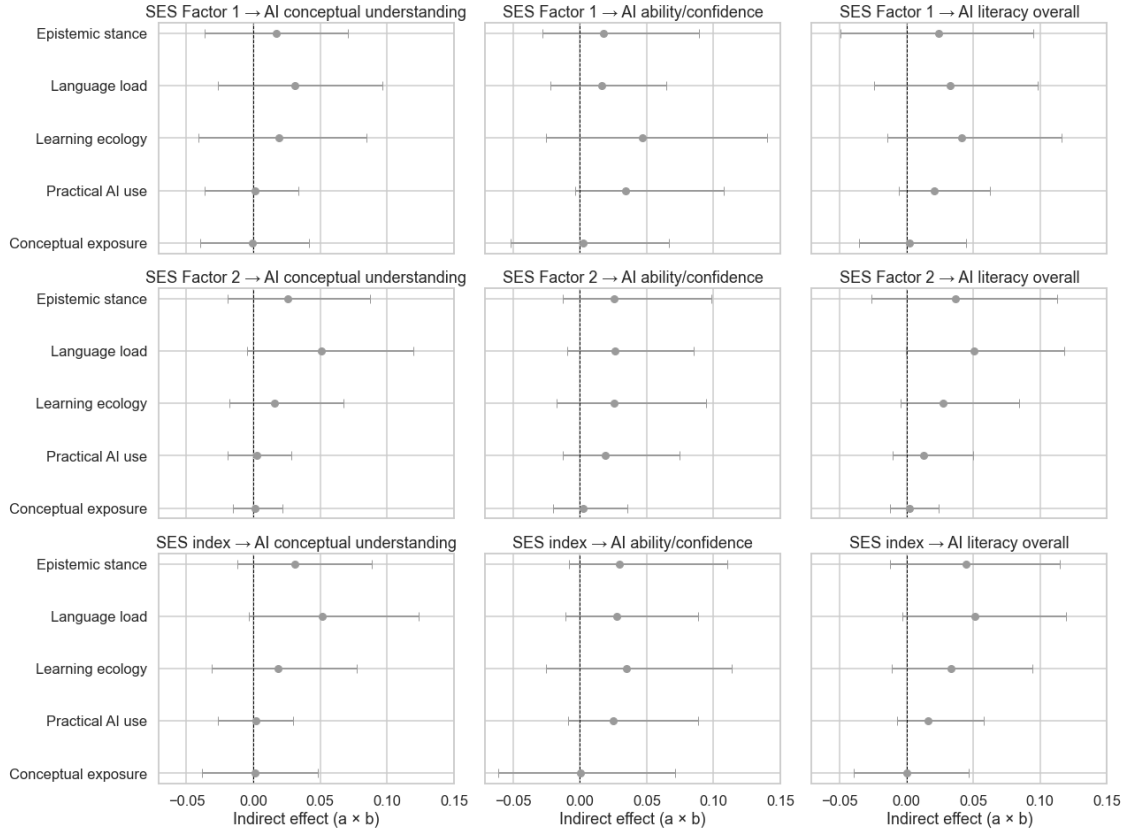
Each point is the estimated indirect effect $a \times b$, and the horizontal line is the bootstrap 95% confidence interval. If the interval stays on one side of zero, that mediator shows clearer evidence of an indirect pathway.

```

[ ]: plot_indirect_effect_forest(
    mediation_results_2ses,
    sample="Combined",
    include_ai_overall=True
)

```

Indirect effects by SES dimension and AI outcome (Combined)



4.3.3 Summary of results (No interaction terms):

1. Conceptual exposure: The closest pathway was

- material household resources → conceptual exposure → overall AI literacy.

coefficient_beta: 0.003, prop_ab_positive = 0.596

The indirect effect was extremely small and unstable, so this does not provide meaningful evidence of mediation. In practice, conceptual exposure does not appear to be an important channel linking SES to AI literacy.

2. Practical AI use: The closest pathway was

- family educational capital → practical AI use → AI ability/confidence.

coefficient_beta: 0.035, prop_ab_positive = 0.946

Students from stronger educational backgrounds may be somewhat more likely to engage in practical AI use, which may in turn support their AI confidence, but the evidence is still not strong enough to claim a confirmed mediation effect.

3. Learning ecology: The closest pathway:

- material household resources → learning ecology → overall AI literacy.

coefficient_beta: 0.027, prop_ab_positive = 0.936

With stronger material and household resources may have somewhat richer learning environments, which may then support AI literacy. However, the indirect effect still falls short of clear statistical support.

4. Language load: The closest pathway:

- material household resources → language load → overall AI literacy.

coefficient_beta: 0.051, prop_ab_positive = 0.974

Students with better material and household resources may experience lower language burden, and lower language burden may in turn support stronger AI literacy. This is the most promising one.

5. Epistemic stance: The closest pathway:

- overall SES → epistemic stance → AI ability/confidence.

coefficient_beta: 0.030, prop_ab_positive = 0.895

Students with higher SES may develop a somewhat stronger critical stance toward technology, which may then support their confidence in handling AI. However, the evidence remains borderline rather than definitive.

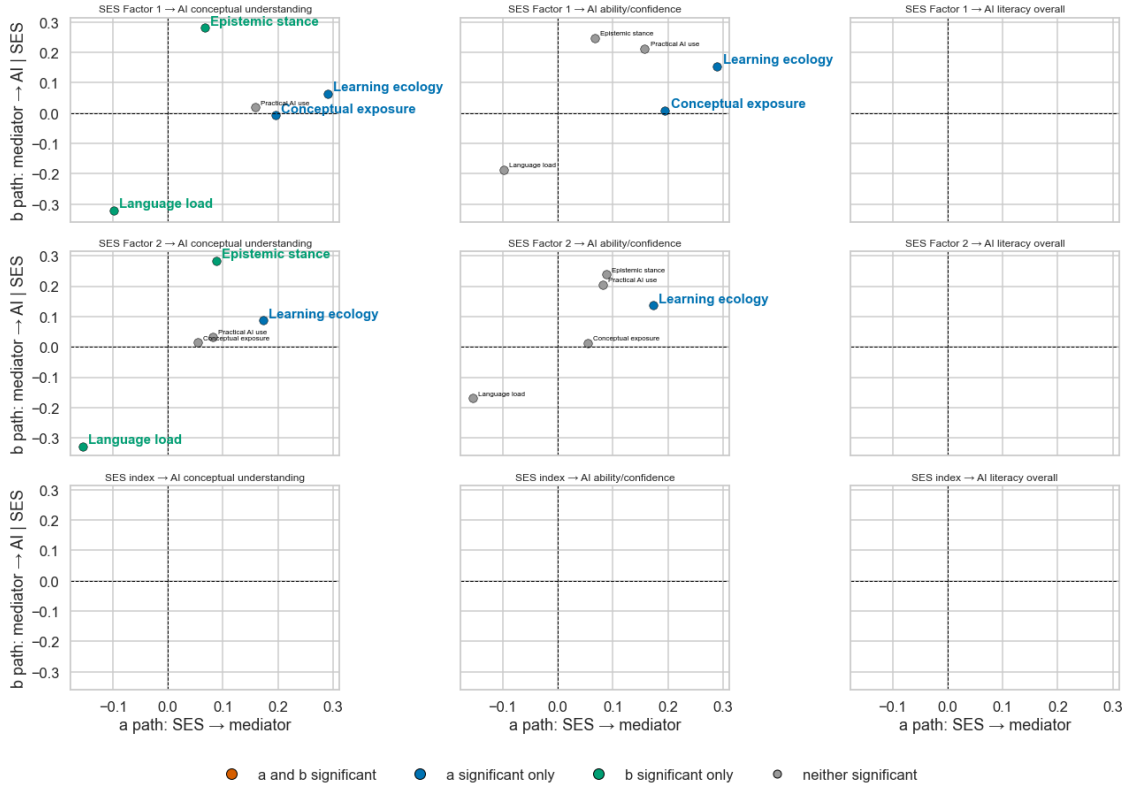
- Overall interpretation: Among the five mediators, language load remains the strongest candidate pathway, with practical AI use, learning ecology, and epistemic stance appearing only as weak secondary candidates. Conceptual exposure does not show a credible mediation role in the current results.

4.3.4 a–b path plot

We need both paths to get a meaningful indirect effect, because mediation only exists when SES is linked to the mediator and the mediator is linked to AI literacy.

```
[ ]: plot_a_b_paths(mediation_results_2ses, sample="Combined")
```

a-b path diagnostic plot (Combined)



4.4 Interaction effects

For two mediators M_1 and M_2 , the interaction model you fitted can be written as:

$$M_{1i} = \alpha_{10} + \alpha_{11}X_i + \varepsilon_{1i}$$

$$M_{2i} = \alpha_{20} + \alpha_{21}X_i + \varepsilon_{2i}$$

$$Y_i = \beta_0 + \beta_1X_i + \beta_2M_{1i} + \beta_3M_{2i} + \beta_4(X_iM_{1i}) + \beta_5(X_iM_{2i}) + \beta_6(M_{1i}M_{2i}) + \varepsilon_i$$

where: - $X_i = SES$ - M_{1i}, M_{2i} = two mediators - Y_i = AI literacy outcome

4.5 Interaction between Mediators

“ses_factor1_score”, “ses_factor2_score”, “ai_factor1_score”, “ai_factor2_score”, “ses_index”, “ai_lit_score”

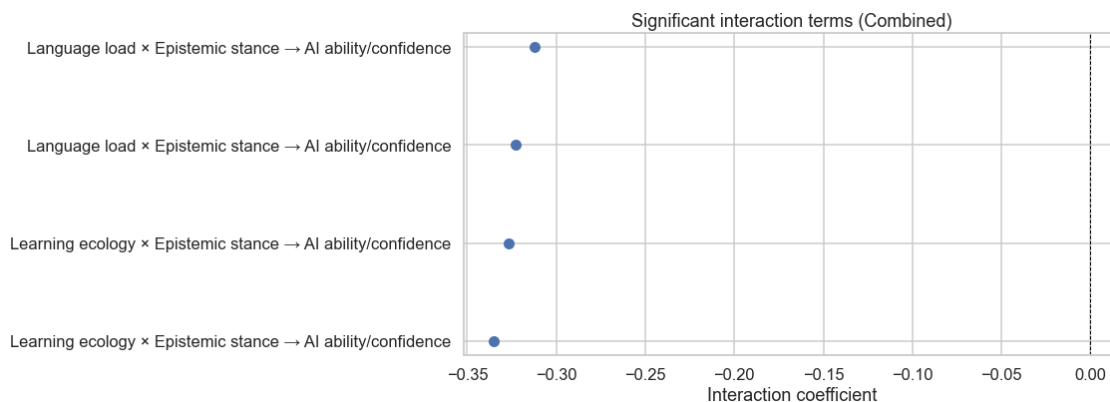
```
[ ]: medi_medi_results = run_interaction_mediation_models(
    df_analysis,
    ses_cols=["ses_factor1_score", "ses_factor2_score"],
    mediators=["learning_ecology_score", "language_load_score",
    ↪ "epistemic_stance_score"],
```

```

    outcomes=["ai_factor1_score", "ai_factor2_score"],
    sample="Combined",
    interaction_type="mediator_mediator"
)

plot_significant_mediator_interactions(
    medi_medi_results,
    alpha=0.05,
    sample="Combined"
)

```



4.6 Interaction between SES and mediators

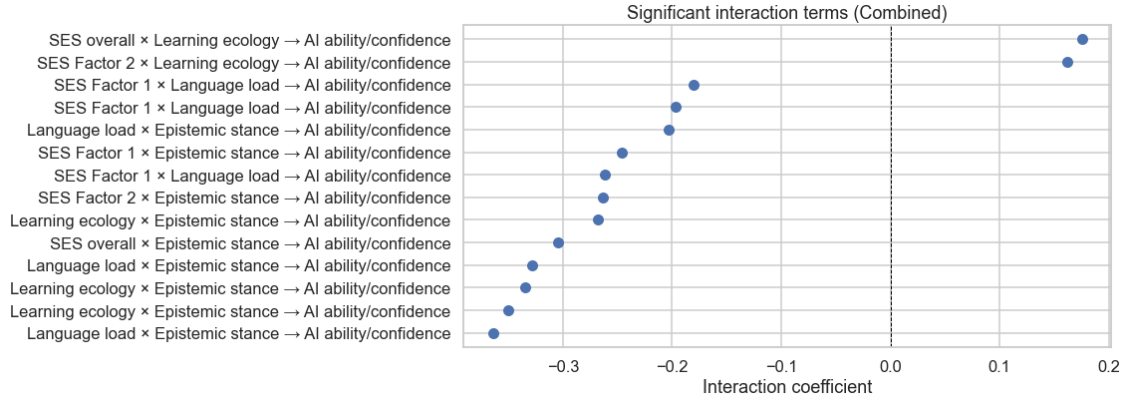
```

[ ]: both_results = run_interaction_mediation_models(
    df_analysis,
    ses_cols=["ses_index", "ses_factor1_score", "ses_factor2_score"],
    mediators=["practical_ai_use_score", "learning_ecology_score",
    ↪ "language_load_score", "epistemic_stance_score"],
    outcomes=["ai_factor1_score", "ai_factor2_score", "ai_lit_score"],
    sample="Combined",
    interaction_type="both"
)

summarize_interaction_mediation_results(both_results, interaction_only=True).
    ↪ round(3)

plot_significant_mediator_interactions(
    both_results,
    alpha=0.05,
    sample="Combined"
)

```



SUMMARY

1. Interaction between Mediators:

1.1. The clearest interaction appears for AI ability/confidence

1.2. Strongest pathway

SES → learning ecology × epistemic stance → AI ability/confidence.

- Interpretation:

Across all three SES definitions, the interaction between learning ecology and epistemic stance is negative and significant. If a student already has a very strong learning ecology, having a stronger epistemic stance adds less extra benefit, and vice versa.

1.3. Secondary pathway:

SES → language load × epistemic stance → AI ability/confidence.

- Interpretation:

A stronger epistemic stance may help students handle AI better, but that benefit becomes weaker when language burden is high. So even if students are thoughtful and reflective about AI, heavy language difficulty may still limit how confident they feel using it.

2. Interaction between SES and mediators:

2.1. The clearest SES × mediator interaction story is also about AI ability/confidence.

2.2 Learning ecology works better at higher SES

2.3 Epistemic stance and language load tend to matter less at higher SES, especially when SES is split into educational and material dimensions

- Interpretation:

Higher-SES students may rely less on certain personal coping mechanisms, like epistemic stance or managing language burden, because their background already provides support. By contrast, supportive learning ecology seems to become even more beneficial when students also have stronger SES resources.